upGrad

Proposal & Intake Form

Bring Your Own Capstone

How your capstone is approved — complete every section

This form will be used to assess your proposal for acceptance or rejection — complete every section. Vague or incomplete proposals are returned for revision, and once accepted your problem and dataset are locked (scope can be scaled down later, but no switching domain or dataset).

1 · Learner & project

Learner name	Aditi Kulkarni
Email ID	Aditi.dilip.kulkarni@outlook.com
Program and track	Masters in AI ML | Gen AI Track | C76 May'25 Batch
Proposed project title	**Last Mile Delivery Optimization for India Multi-Partner Supply Chain**

2 · Problem & business value

Criterion 1 — a measurable organisational problem, not theoretical curiosity.

Business problem / pain point	Last-mile delivery — the final leg of a product's journey from a distribution hub to the customer's door — accounts for 30–53% of total logistics costs and is the single largest driver of customer dissatisfaction in e-commerce. In India, where the logistics network spans five major geographic regions, multiple delivery modes (Standard, Two-Day, Express, Same-Day), multiple delivery partners and highly variable weather conditions – predicting and mitigating delivery delays at scale requires intelligent automation. Operations teams typically respond to delays reactively: a delivery misses its SLA window, a customer calls support, and only then does the team investigate. This reactive model is costly, customer-damaging, and operationally inefficient. A proactive, AI-driven system that can flag risk before dispatch — and suggest remediation automatically — represents a significant competitive advantage. **Business Challenges:** A fictitious company operates a multi-partner last mile delivery network in India, where orders are dynamically allocated to different third-party logistics providers across cities and zones. • As volumes grow, the operations team struggles to predict which orders are likely to breach SLA (e.g., delayed deliveries, first-attempt failures) early enough to intervene. • Delay drivers such as weather patterns, partner capacity, vehicle stress, routes, and geo-specific bottlenecks are not consistently captured, analyzed, or fed back into planning, leading to repeated issues and reactive firefighting. • Existing dashboards are descriptive (showing what already happened) but do not provide forward-looking simulations or prescriptive recommendations on what actions to take today to protect tomorrow's delivery performance and what actions to take in a long-term for delivery optimization. • There is no unified system that connects predicted and simulated delays with codified SOPs, SLAs, and policies, so recommendations are often inconsistent, depend on individual experience, and may violate contractual or policy constraints. • Management lacks a pattern-level, long-term view of recurring problems by lane, partner, city, or customer segment, making it difficult to design structural improvements (e.g., partner mix changes, zone reconfiguration, process changes) based on data.
Why it matters strategically?	Every 1 in 4 daily shipments is delayed. Last-mile delivery accounts for 30–53% of total logistics costs and is the single largest driver of customer dissatisfaction in e-commerce. Without any AI powered automated system an operations analyst spends approx. 30+ minutes per incident manually querying partner data, cross-referencing SLA documents, and drafting communications. "**Last Mile Delivery Optimization for India Multi-Partner Supply Chain**" **project**, will build an Multi-Agentic control tower for last mile operations across multiple logistics partners in India. The system focuses on predicting delivery delays, simulating "what-if" scenarios, diagnosing operational root causes, recommending both short-term quick

	actions and long-term structural improvements to enhance delivery timelines and proactively informing customers thus managing customer experience. The project is structured around two closely integrated sub-systems: • **Machine Learning prediction pipeline** – for delay prediction, delay severity prediction, "what-if" simulations and storing trends and patterns. • **Supply chain delivery pipeline** – Multi-agents workflow for the Operations teams to orchestrate across five output domains: predictions, root-cause diagnosis, what-if simulations, optimization recommendations, and customer email alerts Taken together, the system enables logistics operations teams to query a conversational interface in natural language and receive structured, data-driven intelligence across five output domains.
Earlier solution attempts and pitfalls	• Existing dashboards provide descriptive analytics (delay rates, partner performance tables) but no predictive, prescriptive or recommendation capabilities. • Manual data inference analysis and SLA impact assessment processes are inconsistent and person-dependent. • Ad-hoc rule-based thresholds (e.g., flag same-day orders in rainy/stormy weather) fail to account for compounded and corelated risk signals such as high vehicle load strain combined with tight schedule windows. • Generic LLM chatbot experiments produced non-grounded recommendations not anchored to specific SLA clauses, making them operationally unusable.
Success metric	• **Prediction Agent performance targets**: System should proactively flags at least 4 in 5 at-risk shipments before SLA breach and achieve recall ≥ 80% and ROC-AUC ≥ 90% on delayed orders. Delay severity classifier should achieve weighted F1 ≥ 60% across the three imbalanced severity classes (Short / Medium / Long). • **Diagnosis & Recommendation Agents output quality**: Diagnosis Agent should produce ≥ 5 summarized insights, analyzing the historical patterns (as baseline) and comparing current days order predictions. Recommendation agent should produces ≥ 5 actionable recommendations across short-term, mid-term and long-term categories. Retrieval grounding-- RAG pipeline should show context faithfulness ≥ 0.8. • **Operational efficiency**: End-to-end pipeline (predict → diagnose → simulate (optional) → recommend → email alerts) should complete in ≤ 15 minutes for a 5,000-order daily batch, making it viable as a daily pre-dispatch workflow. • **Business outcome proxy**: System should correctly surface the highest-risk delivery combinations (e.g., Express + Stormy weather, high vehicle load strain on tight schedule windows) with LLM generated auto-inferences and recommendations.
Primary stakeholders	• Logistics Operations Teams (primary daily users: delay prediction and quick-action recommendations) • Supply Chain Managers (strategic: pattern diagnosis, partner SLA review conversations)

3 · Data — availability & legal provenance

Criterion 2 — you must already have the data. "To be sourced later" is rejected.

Dataset name & source	#1. Delivery Logistics Dataset (India – Multi-Partner) ***Source: Kaggle*** | *Author: kundanbedmutha* #2. Partner and Customer SLA Document Custom generated document, with dummy data using AI tools
Link / how it is attached	https://www.kaggle.com/datasets/kundanbedmutha/delivery-logistics-dataset-india-multi-partner Downloaded Raw CSV (Delivery_Logistics.csv) committed to project repository at prediction_pipeline/data/raw/ and included in the submitted GitHub ZIP archive.
How you obtained it	Downloaded directly from Kaggle public repository.
Licence & legal status	Public Kaggle dataset released under the Kaggle Community “Attribution 4.0 International” license. https://creativecommons.org/licenses/by/4.0/ *You are free to:* ***Share*** *— copy and redistribute the material in any medium or format for any purpose, even commercially.* ***Adapt*** *— remix, transform, and build upon the material for any purpose, even commercially. The licensor* ***cannot revoke*** *these freedoms as long as you follow the license terms.*
PII / IP check	**Delivery_Logistics.csv Dataset** is synthetically generated (confirmed by Kaggle dataset author): no real customer names, addresses, phone numbers, order IDs, or GPS coordinates. Delivery partner names (Delhivery, Blue Dart, Ekart, DHL, FedEx, etc.) are publicly known brand labels used as categorical values only. No PII. No proprietary, confidential, or competitively sensitive business data. No IP concerns. **SLA Document** is synthetically generated using AI tools, by me. No proprietary, confidential, or competitively sensitive business data. No IP concerns.
Data format and size	**Delivery_Logistics.csv**: The primary training dataset contains 25,000 historical delivery records representing operations across India's five geographic regions and four delivery modes. The dataset reflects realistic patterns in last-mile logistics, including seasonal weather variation, partner performance variability, operational conditions, package characteristics, environmental factors, and delivery outcomes. The data includes major logistics providers such as Delhivery, Blue Dart, Ekart, DHL, FedEx, Shadowfax, XpressBees, Amazon Logistics, and Ecom Express. Each record describes key delivery attributes including package type, vehicle used, delivery mode, weather condition, travel distance, package weight, and cost estimation. Performance-related fields such as actual delivery time, expected time, delay status, and final delivery status provide insights into how real-world constraints like traffic, vehicle type, and weather influence delivery efficiency. Additionally, a delivery rating is included to reflect customer feedback, helping explore service quality patterns. The dataset is ideal for studying logistics operations, delay analysis, delivery cost behavior, route efficiency, environmental impact, and overall supply chain performance. Since the dataset is synthetically generated, it contains no personal information and is safe for academic, analytical, or business-oriented research. **SLA Document**: A structured Markdown document (delivery_sla_github_ready.md) serves as the organization's SLA policy reference. It contains 36 detailed sections covering performance targets by delivery mode, delay severity thresholds, weather-specific risk profiles, critical combination rules, partner benchmarks, and escalation procedures.

☒ I confirm I already have access to this data now — it is not “to be sourced later.”

4 · Scope & solution approach

Criterion 4 — must meet your track Scope Specification benchmarks.

<table>
<tr><td>Scope-Spec stages & components you will cover</td><td>

“**Last Mile Delivery Optimization for India Multi-Partner Supply Chain**”, builds an AI-assisted control tower for last mile operations across multiple logistics partners in India. The system focuses on predicting delivery delays, simulating “what-if” scenarios, diagnosing operational root causes, recommending both short-term quick actions and long-term structural improvements to enhance delivery timelines and proactively informing customers thus managing customer experience.

The project is structured around two closely integrated sub-systems:

- **Machine Learning prediction pipeline** – for delay prediction, delay severity prediction, “what-if” simulations and storing trends and patterns in an SQLite database.
- **Supply chain delivery pipeline** – Multi-agents workflow along with an interactive UI for the Operations teams to orchestrate across five output domains: predictions, root-cause diagnosis, what-if simulations, optimization recommendations, and customer email alerts

Taken together, the system enables logistics operations teams to query a conversational interface in natural language and receive structured, data-driven intelligence across five output domains.

Machine Learning prediction pipeline will include:

- **Delay Prediction & Delay Severity Prediction ML Classification models**, before or at dispatch time, providing the delivery orders that are at risk of delay and estimate the likely delay severity.
- **Delay historical and current day information Structured Database** which will hold baselines across various operational dimensions, and current day data.

Supply chain last-leg delivery pipeline will include:

Seven specialized AI agents (Predict, Diagnose, Simulate, Recommend, Email, Fallback, and a Master Orchestrator) collaborate sequentially to execute full-pipeline analyses.

- An **AI master agent** that orchestrates specialized sub-agents for delay prediction, scenario simulation, root-cause diagnosis, recommendations, customer alerts – all exposed through a single conversational UI interface for Operations users.
- A **prediction agent** that in turn uses the ML pipeline prediction python tool to score open orders for expected delay, predict the delay severity (hours range) risk, returning prioritized lists that highlight at-risk shipments along with LLM generated inference per shipment for proactive action.
- A **diagnosis agent** that in turn uses the Database query tool and analyzes recent performance against historical baselines to identify dominant root causes of delays (e.g., weather pattern connection, partner under-performance, etc.) at both root-cause and geo levels, and surfaces them in structured tables.
- A **recommendation agent** that compares current batch delay prediction inferences with the ML pipeline historical baselines database query tool along with the SLA policies documents used as a RAG tool to:

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	◦ Generate short-term “quick actions” (rerouting, partner switching, vehicle change, etc). ◦ Provides medium-term and long-term, pattern-based recommendations (partner rebalancing, process changes, policy tuning, etc) using trend analysis and long-horizon improvement playbooks. • A **simulation agent** (optional scope) that allows users to test “what-if” scenarios (e.g., heavy rain in a region, change of vehicle type, etc) using Database query tool with historical patterns and current conditions, and estimates the resulting impact on delivery times. • A **customer alert agent** (optional scope) that generates personalized email templates based on the delay type, reasons and can be triggered for auto-emailing to manage customer experience. • A **fallback agent** (optional scope) which gets invoked if the main pipeline agents fails to deliver. Fallback agent provides high level information and next steps the user can take to circumvent the error and proceed. • A **conversational AI interface** packages all of the above into an accessible interface for a non-technical operations staff to use easily, along with a human-in-a-loop fashion for the workflow orchestration.
Tools / stack (open-source, local) *<u>NOTE:</u> upGrad is not liable to provide access or support for tools or technologies*	• **Data pipeline code**: Python • **Configuration**: python-dotenv, YAML • **EDA & Visualization**: matplotlib, seaborn • **ML model**: sklearn • **Training data**: Kaggle • **Structured Database**: SQLite • **MCP server**: FastMCP • **Vectorstore, RAG**: ChromaDB • **Agents Orchestration Framework**: OpenAI Agents SDK • **Structured input / output**: Pydantic • **LLM backbone**: GPT 4.x, 5.x • **RAG text-chunking**: Langchain-text-splitters • **LLM for embedding**: OpenAI text-embedding-3-small • **Code-encoder reranker**: Sentence-transformers (cross-encoder/ms-macro-MiniLM-L6-v2) • **User Interface**: Gradio • **Package Manager**: uv / uv.lock + requirements.txt • **Testing**: Pytest • **Evaluation**: RAGAS, LLM-as-a-judge All tools and LLM APIs are available with me with personal subscription.

5 · Week-by-week plan

Criterion 3 — a realistic plan fitting ~30–40 hours over four weeks.

Week	Milestone / deliverable	Est. hours
Week 1	**Data Extraction & EDA, ML Models, Database Load**: Load and explore 25,000-record Kaggle dataset; perform EDA; feature engineering; train classifier model for Stage 1 binary delay prediction; train Stage 2 severity classifier model; persist both models; **Database**: design and populate SQLite database with historical data patterns. **MCP Server**: Implement FastMCP server registering 3 ML tools for delay prediction, delay severity prediction, database queries for historical and daily patterns. **Gradio UI**: Conversational front-end, tabs for prediction, diagnosis, recommendation, simulation, email alerts.	15-20
Week 2	**Prediction agent**: MCP client; integration with ML delay prediction tool. LLM to derive inferences per record; display results on UI. LLM to summarize overall results. **Diagnosis agent**: MCP client; integration with Database Query tool. LLM for root cause analysis using historical benchmarks and current batch data comparison; and summarize overall results. **Recommendation agent**: Chunking of SLA policy documents; RAG vectorstore; read output from Prediction and Diagnosis agents combined with SLA retrievals to provide short-term and long-term recommendations. **Gradio UI**: Invoke agents based on natural-language conversation. Integrate outputs with the UI components.	15-20
Week 3	**Simulation agent (optional)**: MCP client; integration with Database Query tool to get historical risk patterns. Gradio UI conversation to translate simulation query as input to Agent tool; display results on UI. LLM to summarize overall results. **Email alert agent (optional)**: integration with function tool to generate email templates for customer intimation. **Fallback agent (optional)**: Simple agent to understand error if any of the previous workflow fails and provide generic steps to retry or abort gracefully. **Gradio UI**: Integrate all code and components through UI.	15-20

	Evaluation & End-to-end Testing: Complete agents and RAG evaluation using LLM-as-a-judge and RAGAS.	
Week 4	**Environment Cleanup**: Project folder structure, packaging, Github Repository + README, LICENSE, requirements.txt, etc. **Documentation**: Technical documentation, Executive summary powerpoint.	15-20

6 · Deliverables

You will submit all three artifacts on the upGrad platform:

☒ Technical report (10–15 pages)

☒ Executive presentation (8–12 slides)

☒ ZIP / GitHub repository + README

7 · Feasibility self-check

☒ The scope realistically fits ~30–40 hours over four weeks.

☒ My data is in hand now and legally usable.

☒ My project meets the "minimum to be accepted" in my Scope Specification.

☒ I have a primary measurable success metric with supporting secondary metrics.

☒ This is an original problem — not a cloned tutorial or a toy dataset.

8 · Declaration

☒ This is my original work. ☒ My data is legally usable. ☒ Scope fits ~30–40 hours over four weeks.

Signature: Aditi Kulkarni Date: 23-June-2026